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FCI

Example

$$|\Psi_0\rangle = |\Phi_0\rangle + \sum_{ai} c_i^a |\Phi_i^a\rangle + \sum_{\substack{ab \\ ij}} c_{ij}^{ab} |\Phi_{ij}^{ab}\rangle + \cdots + \text{NpNh}$$

$$\langle \Phi_0 | \Psi_0 \rangle = 1$$

$$E_0 = \langle \Psi_0 | \hat{H} | \Psi_0 \rangle$$

$$E_0 = E_0^{\text{Ref}} + \Delta E$$

$$E_0^{Ref} = \langle \Phi_0 | H | \Phi_0 \rangle$$

$$\Delta E = \sum_{ia} c_i^a \underbrace{\langle i | f | a \rangle}_{HF = 0}$$

$$+ \sum_{\substack{ab \\ ij}} c_{ij}^{ab} \langle ij | v | ab \rangle_{AS}$$

Many-body pert theory (MBPT)
and Rayleigh-Schrödinger
(RSPT)

$$\Delta E^{(2)} = \sum_{ia} \frac{c_i^a \langle i | f | a \rangle \langle a | f | i \rangle}{\varepsilon_i - \varepsilon_a} +$$

$$+\frac{1}{4}\sum_{\substack{ab\\ij}} \frac{\langle ij|v|ab\rangle\langle ab|v|ij\rangle}{\varepsilon_i+\varepsilon_j-\varepsilon_a-\varepsilon_b}$$

$$c_{ij}^{ab}$$

Example

$$\hat{H}_0 = \sum_{p=1}^{2} \varepsilon_p q_p^+ q_p$$

$$\hat{H}_0|1\rangle = \varepsilon_1|1\rangle$$

$$\hat{H}_0|2\rangle = \varepsilon_2|2\rangle$$

$|2\rangle$ —— ε_2

$|1\rangle$ —— ε_1

$\langle 1|2\rangle = 0$

$\langle i|j\rangle = \delta_{ij}$

$\hat{P} = |1\rangle\langle 1|$, $\hat{Q} = |2\rangle\langle 2|$

$\hat{P}\cdot\hat{Q} = 0$, $\hat{P}^2 = \hat{P}$, $\hat{Q}^2 = \hat{Q}$

$|1\rangle = a_1^+ |0\rangle$ ∧ $|2\rangle = a_2^+ |0\rangle$

$H_1 = g \sum_{pq} a_p^+ a_q$

$\langle 1|H|1\rangle = \varepsilon_1 + g$

$\langle 2|H|2\rangle = \varepsilon_2 + g$

$\langle 1|H|2\rangle = g\langle 1| \sum_{pq} a_p^+ a_q |2\rangle$

$= g \sum_{pq} \langle c| a_1 a_p^+ a_q a_2^+ |c\rangle$

$= g$

Hamiltonian matrix

$$H = \begin{bmatrix} \varepsilon_1 + g & g \\ g & \varepsilon_2 + g \end{bmatrix}$$

$$\det(H - \lambda I_{2\times 2}) = 0$$

$$(\varepsilon_1 + g - \lambda)(\varepsilon_2 + g - \lambda) - g^2 = 0$$

$$\varepsilon_1 + g - \lambda = \frac{g^2}{\varepsilon_2 + g - \lambda}$$

(Brillouin-Wigner PT)

$$\lambda_{1,2} = \frac{\varepsilon_1 + \varepsilon_2}{2} + g \pm \sqrt{\frac{(\varepsilon_1 - \varepsilon_2)^2}{4} + g^2}$$

$$= \frac{\varepsilon_1 + \varepsilon_2}{2} + g \pm (\varepsilon_1 - \varepsilon_2)\sqrt{1 + \frac{4g^2}{(\varepsilon_1 - \varepsilon_2)^2}}$$

$$x = \left(\frac{2g}{\varepsilon_1 - \varepsilon_2}\right)^2 \qquad (|x| < 1)$$

$$\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} + \cdots$$

Lowest eigenvalue

$$\lambda_1 = \varepsilon_1 + \boxed{g} - \frac{g^2}{\varepsilon_1 - \varepsilon_2} + \frac{g^4}{(\varepsilon_1 - \varepsilon_2)^3} + \cdots$$

1st order — 2nd order — 4th-order

Formal derivation of MBPT

$$|\Psi_0\rangle = |\Phi_0\rangle + \sum_{pH} c_H^P |\Phi_H^P\rangle$$

simplified

$$|\Phi_0\rangle + \sum_{m=1}^{\infty} c_m |\Phi_m\rangle$$

$$\hat{H}_0 |\Phi_m\rangle = \varepsilon_m |\Phi_m\rangle$$

$$\hat{H}_0 |\Phi_0\rangle = \varepsilon_0 |\Phi_0\rangle$$

$$H|\Psi_0\rangle = E_0 |\Psi_0\rangle$$

$$\langle \Psi_0 | \Phi_0 \rangle = 1$$

$$\langle \Phi_0 | H | \Psi_0 \rangle = E_0$$

$$\langle \Phi_0 | H | \Phi_0 \rangle = E_0^{Ref}$$

$$E_0 - E_0^{Ref} = \Delta E$$

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$$\langle \Phi_0 | H_0 + H_1 | \Psi_0 \rangle =$$

$$\underbrace{\langle \Phi_0 | H_0 | \Psi_0 \rangle}_{E_0} + \langle \Phi_0 | \hat{H_1} | \Psi_0 \rangle$$

$$= E_0 \Rightarrow E_0 - \epsilon_0 = \underbrace{\langle \Phi_0 | \hat{H_1} | \Psi_0 \rangle}_{\Delta E\,(MBPT)}$$

$$= \underbrace{\langle \Phi_c | H_1 | \Phi_c \rangle}_{\text{1st-order}} + \sum_m c_m \langle \Phi_c | H_1 | \Phi_m \rangle$$

$$\times | \Phi_m \rangle$$

Define $\hat{P} = | \Phi_c \rangle \langle \Phi_c |$

$$\hat{Q} = \sum_{m \geq 1} | \Phi_m \rangle \langle \Phi_m |$$

$$\hat{P}^2 = \hat{P} \qquad \hat{Q}^2 = \hat{Q}$$

$$[\hat{P}, \hat{Q}] = 0$$

$$| \Psi_0 \rangle = (\hat{P} + \hat{Q}) | \Psi_0 \rangle =$$

$$= |\Phi_c\rangle + \hat{Q}|\psi_c\rangle$$

$$(\hat{H}_0 + \hat{H}_1)|\psi_c\rangle = E_0|\psi_c\rangle$$

introduce a parameter w

$$(w - \hat{H}_0)|\psi_c\rangle = (w - E_0 + \hat{H}_1) \times |\psi_c\rangle$$

Resolvent

$$R(w) = (w - \hat{H}_0)^{-1} \quad \left(\frac{1}{w - \hat{H}_0}\right)$$

$$(w - \hat{H}_0)^{-1}(w - H_0) = \underline{\mathbb{1}}$$

$$|\psi_0\rangle = (\omega - H_0)^{-1}(\omega - E_0 + H_1)$$

$$\times |\psi_0\rangle$$

$$\hat{Q}|\psi_0\rangle = Q(\omega - \hat{H}_0)^{-1}(\omega - E_0 + H_1)|\psi_0\rangle$$

$$\left([Q, H_0] = [P, H_0] = 0 \right)$$

$$\hat{Q}|\psi_0\rangle = \frac{\hat{Q}}{\omega - H_0}(\omega - E_0 + H_1)|\psi_0\rangle$$

Reduced resolvent

$$R_Q(\omega) = \hat{Q}\,(\omega - H_0)^{-1} Q$$

$$= \sum_{m \neq 0} \frac{|\Phi_m\rangle\langle\Phi_m|}{\omega - \varepsilon_m} \qquad (*)$$

$$H_0 |\Phi_m\rangle = \varepsilon_m |\Phi_m\rangle$$

$$R_Q(\omega)\,|\Phi_0\rangle = 0 \qquad \langle\Phi_0|\Phi_m\rangle = 0$$

$$R_Q(\omega)\,|\Phi_m\rangle = \frac{|\Phi_m\rangle}{\omega - \varepsilon_m} \qquad (m \neq 0)$$

$$(\omega - \hat{H}_0)\,|\Psi_0\rangle = (\omega - E_0 + H_1)\,|\Psi_0\rangle$$

$$Q\,|\Psi_0\rangle = R_Q(\omega)(\omega - E_0 + H_1)\,|\Psi_0\rangle$$

$$|\Psi_0\rangle = |\Phi_0\rangle + R_Q(\omega)(\omega - E_0 + H_1)\,|\Psi_0\rangle$$

Solve iteratively by repeated substitution:

$$A(\omega) = R_Q(\omega)(\omega - E_0 + H_1)$$

$$|\Psi_0\rangle = |\Phi_0\rangle + A\,|\Psi_0\rangle$$

$$|\Psi_0^{(1)}\rangle = |\Phi_0\rangle + A\,|\Phi_0\rangle$$

$$|\Psi_0^{(2)}\rangle = |\Phi_0\rangle + A\,|\Phi_0\rangle + A^2\,|\Phi_0\rangle$$

$$|\Psi_c\rangle = \sum_{n=0}^{\infty} \left[R_Q(w)(w - E_0 + H_1) \right]^n |\Phi_c\rangle$$

$$\Delta E = \langle \Phi_c | H_1 | \Psi_c \rangle =$$

$$\sum_{n=0}^{\infty} \langle \Phi_0 | H_1 \left[R_Q(w)(w - E_0 + H_1) \right]^n | \Phi_c \rangle$$

Basis for

(i) Brillouin-Wigner PT (BWPT)

$w = E_0$

(ii) Rayleigh-Schrödinger (RSPT)

$w = E_0$

can we prove (*)

$$(\omega - \hat{H}_0)\,|\Psi_0\rangle = (\omega - E_0 + H_1)\,|\Psi_0\rangle$$

$$\hat{Q} = \mathbb{1} - |\Phi_0\rangle\langle\Phi_0|$$

$$\hat{Q}\,(\omega - H_0)^{-1}\,\hat{Q} = \hat{Q}\,(\omega - H_0)^{-1}$$

$$= \frac{\hat{Q}}{\omega - H_0} = R_{\hat{Q}}(\omega)$$

$\hat{Q}$ commutes with $\hat{H}_0$

$$\hat{H}_0\,\hat{Q} = \sum_{m>1} \hat{H}_0\,|\Phi_m\rangle\langle\Phi_m| = \sum_{m>1} \varepsilon_m\,|\Phi_m\rangle\langle\Phi_m|$$

$$\hat{Q}\hat{H}_0 = \sum_{m\geq 1} |\phi_m\rangle\langle\Xi_m| H_0$$

$$= \sum_m |\phi_m\rangle \left(\hat{H}_0^{+} |\Xi_m\rangle\right)^{+}$$

$$= \sum_m \varepsilon_m^{*} |\Xi_m\rangle$$

$$\varepsilon_m^{*} = \varepsilon_m \Rightarrow$$

$$[\hat{Q}, \hat{H}_0] = 0 \quad \text{same with} \quad [\hat{P}, \hat{H}_0] = 0$$

H_0 as matrix $= \text{diag}(\varepsilon_0, \varepsilon_1, \varepsilon_2 \ldots)$

$\hat{Q} = \mathrm{diag}(0, 1, 1, \ldots)$

Two diagonal matrices, always commute

if $A = w - H_0$

if $\hat{A}$ commutes with $\hat{Q}$, then $\hat{Q}$ commutes with A^{-1}

Proof $\hat{Q}\hat{A} = \hat{A}\hat{Q}$

multiply with A^{-1} from the left and right

$$A^{-1}(QA)A^{-1} = A^{-1}(AQ)A^{-1}$$

$$AA^{-1} = \mathbb{1}$$

$$A^{-1}Q \cdot \mathbb{1} = \mathbb{1}\, QA^{-1} \Rightarrow$$

$$A^{-1}Q = QA^{-1}$$

$$Q(\omega - H_0)^{-1} = (\omega - H_0)^{-1}Q$$

$$\hat{Q}(\omega - \hat{H}_0)^{-1}\hat{Q} = \hat{Q}^2(\omega - \hat{H}_0)^{-1}$$

$$= \hat{Q}(\omega - \hat{H}_0)^{-1}$$

can write as $\dfrac{\hat{Q}}{\omega - \hat{H}_0}$

The spectral form of the resolvent

$$G_0(\omega) = \sum_{n \geq 0} \frac{|\Phi_n\rangle\langle\Phi_n|}{\omega - \varepsilon_n}$$

$$(\omega - \hat{H}_0) G_0(\omega) = \sum_n \frac{(\omega - \hat{H}_0) \times |\Phi_n\rangle\langle\Phi_n|}{\omega - \varepsilon_n}$$

$$\left(H_0 |\Phi_n\rangle = E_n |\Phi_n\rangle \right)$$

$$= \sum_{n \geq 0} |\Phi_n\rangle\langle\Phi_n| = \mathbb{1}$$

The same holds for the factors in the opposite order

$$\frac{1}{\omega - H_0} = \sum_{n \geq 0} \frac{|\Phi_n\rangle\langle\Phi_n|}{\omega - E_n}$$

$$\langle\Phi_m| \frac{1}{\omega - H_0} |\Phi_n\rangle = \langle\Phi_m| (\omega - H_0)^{-1} \times |\Phi_n\rangle =$$

$$= \frac{\delta_{nm}}{\omega - \varepsilon_n}$$

$n = 0 \qquad \omega - \varepsilon_0$ can be

$= 0$ if $\omega = \varepsilon_0$ (RSPT)

multiply with $\hat{Q}$

$$\hat{Q}(\omega - \hat{H}_0)^{-1} = \sum_{n \geq 1} \sum_{m \geq 0} |\Phi_m\rangle \frac{\langle \Phi_m | \Phi_m \rangle}{\omega - \varepsilon_m} \langle \Phi_m | = \sum_{m \geq 1} \frac{|\Phi_m\rangle\langle\Phi_m|}{\omega - \varepsilon_m}$$

or a matrix $\left(\frac{Q}{\omega - H_Q}\right)$

$$Q(\omega - H_Q)^{-1} = A^{-1} = 1/A$$

$$\begin{bmatrix} 0 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\omega - \epsilon_0} & & & \\ & \frac{1}{\omega - \epsilon_1} & & \\ & & \ddots & \\ & & & \end{bmatrix}$$

$$= \begin{bmatrix} 0 & & & \\ & \frac{1}{\omega - \epsilon_1} & & \\ & & \frac{1}{\omega - \epsilon_2} & \\ & & & \ddots \frac{1}{\omega - \epsilon_D} \end{bmatrix}$$

$$= R_Q(\omega)$$

BWPT $\omega = E_0$

$$R_Q(\omega) = \hat{Q} \frac{1}{E_0 - Q H_1 Q} \hat{Q}$$

RSPT $\omega = \varepsilon_0$

$$R_Q(\omega) = \hat{Q} \frac{1}{\varepsilon_0 - Q H_0 Q} \hat{Q}$$

$$\Delta E = \sum_{n=0}^{\infty} \langle \Phi_0 | H_1 \left[\bar{R}_Q(\omega)(\omega - E_0 + H_1) \right]^n \times | \Phi_0 \rangle$$

$$\Delta E^{(1)} = \langle \Phi_0 | H_1 | \Phi_0 \rangle$$

$$\Delta E^{(2)} = \langle \Phi_0 | H_1 R_Q(w) H_1 | \Phi_0 \rangle$$

RSPT

$$\Delta E^{(2)} = \sum_{m \geq 1} \frac{|\langle \Phi_m | H_1 | \Phi_0 \rangle|^2}{\varepsilon_0 - \varepsilon_m}$$

$$R_Q(w) = \sum_{m \geq 1} \frac{|\Phi_m\rangle\langle\Phi_m|}{\varepsilon_0 - \varepsilon_m}$$